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GRAPH CONVOLUTIONAL NETWORKS FOR WEB SERVICE CLASSIFICATION

(Li et al., 2021)

TITLE PAGE

Graph Convolutional Networks for Web Service Classification

A Mini Project Report Submitted in Partial Fulfillment of Literature Survey Requirements

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ABSTRACT

Graph Convolutional Networks (GCNs) are advanced deep learning models designed to process graph-structured data. In web service classification, relationships among services play an important role in improving classification accuracy. Traditional machine learning methods fail to capture hidden dependencies among services, whereas GCN models effectively analyze node connections and feature relationships.

This report discusses the implementation of GCN-based classification systems for quality-based web service classification. The proposed approach improves service recommendation and classification performance through graph learning mechanisms. The report also analyzes advantages, disadvantages, observations, and applications of Graph Convolutional Networks in intelligent service classification systems.

INTRODUCTION

Web services are software applications that communicate over the internet using protocols such as SOAP and REST APIs. Due to the increasing number of web services available online, selecting high-quality services has become difficult.

Traditional classification methods mainly depend on feature values but fail to capture hidden relationships among services. Graph Convolutional Networks overcome this limitation by representing services as graph nodes and their relationships as graph edges.

GCNs are capable of learning complex service dependencies and improving prediction performance. This makes them suitable for intelligent web service classification systems.

OBJECTIVES

Objectives of the Study

1. To study Graph Convolutional Networks.
2. To understand graph-based learning.
3. To classify web services effectively.
4. To improve prediction accuracy.
5. To analyze relationships among services.
6. To compare GCN with traditional classifiers.

EXISTING SYSTEM

Before Graph Convolutional Networks, traditional supervised machine learning algorithms were used for web service classification.

These include:

- Decision Tree
- Support Vector Machine
- Logistic Regression
- Random Forest

Limitations

- Poor relationship learning
- Low scalability
- Weak performance on complex datasets
- High dependency on labeled data

Traditional algorithms could not capture hidden dependencies between services.

PROPOSED SYSTEM

The proposed system uses Graph Convolutional Networks to improve classification performance.

Technique Used

- Graph representation
- Node embedding
- Feature propagation
- Deep graph learning

Each web service is represented as a node. Similar services are connected using graph edges. The network learns hidden patterns from these relationships.

METHODOLOGY

Steps Involved

1. Data Collection
2. Graph Construction
3. Feature Extraction
4. Graph Learning
5. Node Classification
6. Result Evaluation

The QoS dataset is converted into graph format before training the GCN model.

WORKING PROCESS

The system first collects QoS attributes such as:

- Response Time
- Throughput
- Availability
- Reliability

After preprocessing, graph structures are created. GCN layers then propagate information across neighboring nodes.

Finally, services are classified into:

- Platinum
- Gold
- Silver
- Bronze

ADVANTAGES

Advantages of GCN

- Better classification accuracy
- Learns hidden relationships
- Handles large datasets efficiently
- Better generalization
- Improves recommendation quality

DISADVANTAGES

Disadvantages

- Complex implementation
- Requires high computational power
- Difficult parameter tuning
- Large memory usage
- Requires domain expertise

OBSERVATION

The Graph Convolutional Network showed better performance compared to traditional supervised models. The graph-based learning mechanism successfully captured hidden dependencies among services.

Classification performance improved significantly for partially labeled datasets.

RESULT ANALYSIS

Performance Analysis

GCN models achieved:

- Higher precision
- Better recall
- Improved F1-score
- Better scalability

The model handled complex service relationships efficiently.

APPLICATIONS

Applications of GCN

- Web service recommendation
- Cloud computing
- AI systems
- Smart service platforms
- E-commerce applications

CONCLUSION

Graph Convolutional Networks provide an advanced and intelligent solution for web service classification. By learning hidden relationships among services, GCN improves prediction accuracy and scalability.

The study demonstrates that graph-based learning is more efficient than traditional machine learning approaches for complex service classification problems.

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